

MENGQI LEI

+1 (614) 463-0752 ✉ simonlei9889@gmail.com

🌐 <https://www.linkedin.com/in/mengqi-lei/> 🐙 <https://github.com/MqLei423>

Education

University of Illinois Urbana-Champaign

Champaign, IL

Master of Computer Science; GPA: 4.0/4.0

Aug 2025 – May 2027

The Ohio State University

Columbus, OH

B.S. in Computer Science and Engineering; GPA: 3.6/4.0, Cum Laude

Aug 2021 – May 2025

- Selected 5 times for the College of Engineering Dean's List

Technical Skills

Programming Languages: Java, Python, C, C++, C#, Kotlin, JavaScript, TypeScript, SQL, Assembly, HTML, CSS

Frameworks/Tools: Android Studio, Unity, React.js, Node.js, Next.js, Firebase, Git, Visual Studio, VS Code

Cloud & DevOps: Vercel, AWS, Docker, CI/CD, Linux, Version Control, Kubernetes, Agile Development

Databases: PostgreSQL, SQLite, MySQL, MongoDB, Room, Prisma ORM

Experience

PC Game Developer Intern

Apr 2024 – Jun 2025

Resilience, Inc

Remote

- Designed and implemented 2D and 3D gameplay systems in **Unity** using **C#**, including player mechanics, physics interactions, and interactive UI components to enhance user engagement and visual responsiveness.
- Built gameplay logic modules and optimized in-engine systems to support scalable interactions and event handling, ensuring smooth communication between user inputs and game state.
- Deployed playable builds on **itch.io** for internal testing and public showcasing, demonstrating full development lifecycle ownership from prototype to distribution.
- Applied **Agile Scrum methodology** through sprints and stand-ups, maintaining code quality via **PlasticSCM** version control and peer reviews, reducing bug resolution time and accelerating feature delivery.

Teaching Assistant

Aug 2024 – May 2025

The Ohio State University, College of Engineering (CSE3541)

Columbus, OH

- Assisted 180+ students with **Unity** and **C#** projects, clarifying technical concepts and debugging game mechanics.
- Facilitated communication between students and course staff to ensure project progress and timely support.
- Supported curriculum delivery by evaluating assignments and providing detailed technical feedback.

Projects

Flysmart — Next.js, TypeScript, React Leaflet, Turf.js, Tailwind CSS, Prisma, PostgreSQL, AWS

- Developed a full-stack lie-flat business class route visualization platform with a **Next.js**, **TypeScript**, and **React Leaflet** frontend, featuring dynamic map-based exploration, real-time route rendering, and an intuitive UI for visualizing airline networks.
- Designed and implemented the backend using **PostgreSQL** and **Prisma ORM on Neon**, modeling airports, airlines, aircraft, and routes with optimized relational queries and integrated **RESTful APIs** through **Next.js API routes** for efficient data communication.
- Built scalable backend services by integrating **AWS Lambda** for serverless data processing and **AWS ECR** for containerized microservices, enabling modular deployment and performance scalability.
- Deployed the application on **Vercel** with automated **CI/CD via Git integration**, ensuring continuous testing, secure environment configuration, and frictionless updates on every commit.

Carbon Emission Tracker — JavaScript, Leaflet.js, HTML, CSS, GitHub Pages

- Built an interactive web application using **JavaScript** and **Leaflet.js** to visualize travel routes, transportation modes, and estimated CO₂ emissions with dynamic map rendering and responsive UI design.
- Modeled emission calculations directly in **JavaScript** to process and aggregate route data in real time, eliminating the need for a server-side component and ensuring lightweight client performance.
- Deployed the static web app on **GitHub Pages**, optimizing file structure and configurations for direct browser accessibility and fast load times.
- Implemented **Git-based version control** and integrated **CI/CD** workflows through GitHub Actions to automate build validation and ensure consistent deployment with every commit.